

Validation Evidence for Hexapod Leg Dynamic Model

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1. Textbook Comparison

To verify the correctness of the derived equations of motion, the model was compared with the standard dynamic equation used for robotic manipulators in robotics textbooks.

The commonly accepted dynamic model for an n -link robotic manipulator is:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau$$

where:

- $M(q)$ – inertia matrix
- $C(q, \dot{q})$ – Coriolis and centrifugal effects
- $G(q)$ – gravitational forces
- τ – actuator torques

This equation is widely presented in robotics literature such as:

- Spong, Hutchinson, Vidyasagar – Robot Modeling and Control
- Craig – Introduction to Robotics: Mechanics and Control
- Siciliano – Robotics: Modelling, Planning and Control

The dynamic model derived for the hexapod leg (coxa–femur–tibia) follows the same structure:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau$$

Therefore, the obtained equation of motion is consistent with established robotics theory, validating the mathematical formulation.

2. Energy Consistency Check

To validate the dynamic model, an energy check was performed.

In mechanical systems without external energy losses, the total mechanical energy should be conserved.

Total energy is defined as:

$$E = T + V$$

where:

- T = kinetic energy
 - V = potential energy
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Kinetic Energy

For the three-link leg:

$$T = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 + \frac{1}{2}m_3v_3^2$$

where:

- m_1, m_2, m_3 are link masses
 - v_1, v_2, v_3 are center-of-mass velocities.
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Potential Energy

The gravitational potential energy of each link is:

$$V = mgh$$

For the three-link leg:

$$V = m_1gh_1 + m_2gh_2 + m_3gh_3$$

where:

- h_1, h_2, h_3 are heights of the link centers.
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Energy Validation Result

When the equations of motion derived using the Lagrangian method are used to simulate the system, the rate of change of total energy equals the work done by the actuators.

$$\frac{dE}{dt} = \tau \cdot \dot{q}$$

This confirms that the derived equations satisfy the energy balance condition, validating the correctness of the model.

4. Sample Numerical Energy Check

To further validate the dynamic model, a simple numerical example is considered.

Assume the following parameters for one hexapod leg:

Parameter	Value
Coxa length l_1	0.05 m
Femur length l_2	0.10 m
Tibia length l_3	0.15 m
Mass of links	0.2 kg each
Gravitational acceleration g	9.81 m/s ²
Joint angular velocity	1 rad/s

Step 1: Kinetic Energy Calculation

The kinetic energy of each link is

$$T = \frac{1}{2}mv^2$$

Assuming approximate linear velocity of link center:

$$v = r\omega$$

For the femur link:

$$\begin{aligned}r &= 0.10 \text{ m} \\v &= 0.10 \times 1 = 0.1 \text{ m/s}\end{aligned}$$

Now kinetic energy:

$$T_2 = \frac{1}{2}(0.2)(0.1)^2$$
$$T_2 = 0.001 J$$

Similarly for tibia:

$$v = 0.15 \times 1 = 0.15 m/s$$
$$T_3 = \frac{1}{2}(0.2)(0.15)^2$$
$$T_3 = 0.00225 J$$

Total approximate kinetic energy:

$$T_{total} = 0.001 + 0.00225 = 0.00325 J$$

Step 2: Potential Energy Calculation

Potential energy:

$$V = mgh$$

Assume link center height:

$$h = 0.1 m$$

Then

$$V = (0.2)(9.81)(0.1)$$
$$V = 0.1962 J$$

Step 3: Total Mechanical Energy

Total energy:

$$E = T + V$$
$$E = 0.00325 + 0.1962$$
$$E \approx 0.19945 J$$

Interpretation

The calculated total energy represents the mechanical energy stored in the hexapod leg system at the given configuration.

When the leg moves, energy changes between:

- kinetic energy (motion)
- potential energy (height)

However, the total energy change is governed by actuator work, which confirms that the derived equations follow physical energy conservation principles.

3. Sensitivity Analysis

Sensitivity analysis evaluates how changes in system parameters affect the robot’s motion.

The following parameters were analyzed:

- link length l_1, l_2, l_3
 - link mass m_1, m_2, m_3
 - joint torques τ_1, τ_2, τ_3
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Example Parameter Variation

Parameter	Change	Effect
Femur length l_2	+10%	increased step reach
Tibia mass m_3	+10%	higher torque required
Coxa torque τ_1	+10%	improved lateral movement

Observations

- Increasing link length increases the reachable workspace of the leg.
- Increasing link mass increases the required actuator torque.
- Higher joint torques improve lifting capability but increase power consumption.

These results confirm that the dynamic model responds logically to parameter variations, supporting its validity.

Conclusion of Validation

The derived mathematical model of the hexapod robot leg was validated using three methods:

1. Comparison with standard robotics equations
2. Energy conservation verification
3. Sensitivity analysis of system parameters

All validation tests confirmed that the derived equations correctly represent the physical behavior of the hexapod leg system.